

Rai Yashmit

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SKILLS

Languages: C++, C, Python, SQL

Tech Stack: Power BI, Excel, Docker, Git

Frameworks: TensorFlow, Scikit-Learn, Fast API, NumPy, Pandas

Core Skills: Data Structures and Algorithm, System Design, Machine Learning, Data Analysis

Soft Skills: Strategic Thinking, Adaptability, Rapid Skill Acquisition, Leadership

PROJECTS

Denoising Diffusion Probabilistic Model (DDPM) for Image Generation on MNIST

Dec' 25

- Constructed a **conditional DDPM** with **1,000 diffusion timesteps**, taught on **60k MNIST images** using TensorFlow.
- Reduced reconstruction loss by **~10%** and maintained a train-validation loss gap **<0.001**, indicating strong **generalization**.
- Achieved **stable convergence with early stopping**, reaching a best **validation MSE of ~0.039**.

Sorting Visualizer | Python, FastAPI, Docker

Jun' 25

- Engineered a backend computation engine to execute and visualize **12+ algorithms (sorting and graph-based)** with step-by-step state outputs.
- Automated delivery of educational metadata for each algorithm, serving **30+ unique data points per request**.
- Implemented optimized data structures and algorithms including Quick Sort, Merge Sort, Heap Sort, Dijkstra's, BFS, DFS, Prism, Kruskal and etc.
- Containerized the backend application using **Docker and docker-compose**, ensuring consistent development and deployment across local and production environments

TRAINING

Library Management System | Python, SQLite, FastAPI

Aug' 25

- Architected a RESTful backend system using FastAPI, handling 20+ endpoints for user, inventory, and transaction management.
- Integrated a user preference engine** supporting multi-attribute subscriptions and implemented an **Observer Pattern-based notification system** for real-time book availability alerts.
- Designed a normalized relational schema (3NF) supporting **5+ entities and complex relationships (1:N, M:N)**.

CERTIFICATES AND CERTIFICATIONS

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|---------------------------------------|---------|
| • Data Science Oracle | Oct' 25 |
| • Data Structures and Algorithm LPU | Jul' 25 |
| • Elite in Cloud Computing NPTEL | May' 25 |

EDUCATION

Lovely Professional University

Phagwara, Punjab

Bachelors of Technology in Computer Science and Engineering

Aug' 23 - Present

CGPA – **8.87**